
Wastewater Treatment Through Multiple CSTRs in Series

CHE 204 -- Chemical Reaction Engineering

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Abstract

This report presents a study on wastewater treatment using multiple Continuous Stirred-Tank Reactors (CSTRs) arranged in series. We plan to treat industrial wastewater containing factory chemicals, textile dyes, and pharmaceutical waste by passing it through a cascade of CSTRs. The working equations for both isothermal and adiabatic CSTR configurations are presented, along with Monod kinetics and the IWA Activated Sludge Model framework for biological treatment. A cost model is developed with fixed capital of Rs. 50,000 per reactor and operating costs of Rs. 10,000 per reactor per year. As the number of CSTRs increases, costs increase, so we need to find the right balance between costs, efficiency and safety. Through marginal cost analysis, we find that $N = 3$ CSTRs in series is the recommended optimum, achieving 94.73% conversion at Rs. 65,000/year. Beyond this, the cost per 1% conversion gain doubles. Aerobic and anaerobic configurations, mixing considerations, environmental regulations (EPA/WHO), and the role of the Damkohler number are also discussed. The key conclusion is that reactor design is an optimisation problem, not a maximisation problem.

Assumptions

The following assumptions have been used throughout this report:

1. The reaction is first-order irreversible with rate $= k * C_A$.
2. Perfect mixing inside each CSTR (uniform composition and temperature).
3. Steady-state operation for all reactors in series.
4. For isothermal CSTRs, temperature is constant across all reactors, so k does not change.
5. For adiabatic CSTRs, no heat exchange with surroundings. Temperature changes based on energy balance.
6. The wastewater treatment reaction is exothermic for adiabatic analysis.
7. Total reactor volume is fixed. Only per-reactor fixed cost and operating cost scale with N .
8. Ideal gas behaviour and constant density where applicable.
9. Monod kinetics parameters (μ_{max} , K_S , Y) are constant for a given microbial population.
10. 90% conversion target used as the regulatory baseline.

Chapter 1

Introduction

1.1 Background and Motivation

Over 80% of the world's wastewater is discharged into the environment completely untreated. That's factory chemicals, textile dyes, pharmaceutical waste -- all flowing into the same rivers and groundwater that communities depend on. This is a serious problem because untreated wastewater contaminates drinking water sources, damages ecosystems, and causes long-term health issues for populations living downstream.

We plan to treat the wastewater by passing it through multiple CSTRs kept in series. The CSTR is a well-known reactor type in chemical engineering -- it provides uniform mixing, steady output quality, and is relatively simple to design and control. When you put several CSTRs in series, each reactor progressively reduces the pollutant concentration, and in the limit of a very large number of reactors, the system starts behaving like a Plug Flow Reactor (PFR).

But as the number of CSTRs increase, the costs increase as well. So we eventually have to find a right balance between costs, efficiency and safety. That is the central question this report tries to answer.

1.2 Objectives

The main objectives of this project are: (a) to present the working equations for CSTRs in series under both isothermal and adiabatic conditions, (b) to compare the efficiency of isothermal and adiabatic configurations using the Arrhenius rate law, (c) to develop a cost model and find the optimal number of CSTRs through marginal cost analysis, and (d) to discuss aerobic vs. anaerobic configurations and environmental regulatory requirements.

Chapter 2

Reaction Kinetics and Working Equations

2.1 Monod Kinetics and ASM Models

Biological wastewater treatment works because microorganisms consume the organic pollutants as substrate. The growth rate of these microorganisms on a substrate S is described by Monod kinetics:

$$\mu = \mu_{max} \cdot \frac{S}{K_S + S}$$

(2.1)

Here μ is the specific growth rate, μ_{max} is the maximum growth rate, S is the substrate concentration, and K_S is the half-saturation constant. The substrate utilisation rate is $r_S = -\mu \cdot X / Y$, where X is biomass concentration and Y is the yield coefficient.

For more detailed modelling, the IWA Activated Sludge Models (ASM1/2/3) include multiple state variables -- soluble and particulate organics, biomass, etc. -- and model processes like hydrolysis, nitrification, and denitrification. ASM1 handles carbon oxidation, nitrification (NH_4^+ to NO_3^-), and denitrification, implemented via cascaded CSTRs.

2.2 CSTR Working Equations -- Isothermal Series

We will primarily be talking about two types of CSTRs -- Isothermal (cleaning at a fixed temperature) and Adiabatic (no energy is released outside the CSTR). In an isothermal reactor, the value of k remains constant and the general mole balance equation gives:

$$V_n = \frac{v_0(C_{A,n-1} - C_{A,n})}{k}$$

(2.2)

The temperature equation is $T_n = T_{(n-1)}$, $k = \text{constant}$, and the mole balance is independent of the energy balance. The overall conversion after N equal-volume CSTRs in series is:

$$X_N = 1 - \prod_{i=1}^N \frac{1}{1 + Da_i}$$

(2.3)

where $Da_i = k \cdot V_i / F$ is the Damkohler number for reactor i. In series, the relaxation times are additive:

$$\tau_{total} = \tau_1 + \tau_2 + \dots + \tau_N, \quad \tau_i = \frac{V_i}{v_0}$$

(2.4)

2.3 CSTR Working Equations -- Adiabatic Series

For an adiabatic reactor, there is no heat entering or released from the reactor. We get two equations -- an energy balance and a mole balance:

$$V_n = \frac{v_0(C_{A,n-1} - C_{A,n})}{k(T_n)}$$

(2.5)

The temperature in each reactor changes according to:

$$T_n = T_{n-1} + \Delta T_{ad}, \quad \Delta T_{ad} = \frac{(-\Delta H_{rxn}) \cdot C_{A0} \cdot X}{\sum C_{pi}}$$

(2.6)

The rate constant is temperature-dependent via the Arrhenius equation, $k_n = f(T_n)$. The mole balance and energy balance are coupled -- we must solve for both X and T together.

Table 2.1: Summary of isothermal vs. adiabatic CSTR equations

Feature	Isothermal Series	Adiabatic Series
Mole Balance	$V_n = v_0(C_{A,n-1} - C_{A,n})/k$	$V_n = v_0(C_{A,n-1} - C_{A,n})/k(T_n)$
Temp. Equation	$T_n = T_{n-1}$	$T_n = T_{n-1} + \Delta T_{adiabatic}$
Rate Constant	$k = \text{constant}$	$k_n = f(T_n)$ (Arrhenius)
Key Coupling	Independent of Energy Balance	Coupled: Must solve for X and T together

Chapter 3

Efficiency Analysis -- Isothermal vs. Adiabatic

We are taking 90 percent conversion as a basis. For a first-order wastewater reaction, the residence time required for conversion X is:

$$\tau = \frac{X}{k(1-X)}$$

(3.1)

The Arrhenius Rate Law governs the temperature-dependent rate constant:

$$k = Ae^{-E_a/RT}$$

(3.2)

3.1 Comparative Conditions

For the isothermal case, T is constant (T₀) and k is fixed:

$$\tau_{iso} = \frac{X}{k_0(1-X)}$$

(3.3)

For the adiabatic case, T increases as the reaction progresses (T > T₀) because the wastewater treatment reaction is exothermic:

$$T = T_0 + \frac{(-\Delta H_{rxn}) \cdot C_{A0} \cdot X}{\sum C_{pi}}$$

(3.4)

$$\tau_{adia} = \frac{X}{k(T)(1-X)}$$

(3.5)

3.2 The Efficiency Verdict

For exothermic cleaning, T_{adia} > T₀, therefore k(T) >> k₀. The efficiency ratio is:

$$\eta = \frac{V_{iso}}{V_{adia}} = \frac{k(T)_{adia}}{k_0}$$

(3.6)

The adiabatic CSTR is often 2x to 5x smaller in volume because the heat of reaction self-accelerates pollutant destruction. However, we prefer isothermal operation because safety is the priority. In an isothermal system, the temperature is an independent variable -- you decide the temperature, and the reaction follows. Moreover, most wastewater treatment is biological and keeping the reactor isothermal is important for microbes to perform the function of cleaning.

Chapter 4

Aerobic vs. Anaerobic CSTR Configuration

4.1 Aerobic CSTR

Setup: Tank with mixing + aeration (using diffusers or surface aerators); usually 1-3 tanks in series. The process involves BOD (Biochemical Oxygen Demand) removal by microorganisms, nitrification (NH_4^+ to NO_3^-), and denitrification (in anoxic conditions: no oxygen but nitrate present). Sludge from the clarifier is returned as RAS (Return Activated Sludge) to maintain biomass and control SRT (Sludge Retention Time).

Advantages: Good removal of BOD and TSS (Total Suspended Solids), handles load changes well, smaller area required. Limitations: High oxygen cost, more sludge production, sensitive to toxic chemicals.

4.2 Anaerobic CSTR

Setup: Closed tank, no oxygen, mixing by mechanical stirrer or recirculation (often heated). The process follows: Hydrolysis, then Acidogenesis, then Acetogenesis, then Methanogenesis. This produces biogas (mainly methane CH_4). Design requires long HRT (Hydraulic Retention Time) of 10-30 days; often pre-treatment like DAF (Dissolved Air Flotation) is used.

Advantages: Treats high COD (Chemical Oxygen Demand) waste with about 80% removal, produces energy (biogas), low sludge generation. Limitations: Sensitive to pH, slow startup, needs heating in cold conditions, does not remove nutrients (Nitrogen and Phosphorus).

Key Difference: Aerobic = needs oxygen, faster, cleaner effluent. Anaerobic = no oxygen, slower, produces energy (biogas).

4.3 Mixing and Scale-Up

Mixing is essential to maintain uniform conditions (nutrients, oxygen). It is achieved by mechanical agitators or diffused aeration (air bubbles provide mixing). Typical aerobic aeration power is about 0.5-2 kW per 1000 m³ depending on design. Overmixing wastes energy. Scale-up rules: Maintain similar mixing regime (Reynolds number, tip speed). Larger tanks may use multiple mixers/aerators. For aerobic CSTR, design k_La (volumetric transfer coefficient) to meet OUR (oxygen uptake rate).

4.4 Advantages of CSTRs

The main advantages of CSTRs for wastewater treatment are: uniform treatment (steady output quality), simple design and control (well-studied models), handles sudden load variations (due to complete mixing), and can process high solids (effective stirring and mixing).

Chapter 5

Cost vs. Conversion Analysis

Under this section we aim to find the right balance between cost and conversion. We try to view how much effect adding a CSTR provides and whether it is worth the cost. We can try to keep multiple CSTRs in order for the function to act like a PFR but it won't be cost effective.

5.1 Cost Model

Each reactor has two cost components. Capital: Rs. 50,000 (fixed per reactor) + Rs. 2,000/L (volume-dependent). Operating: Rs. 10,000/reactor/year (mixing, maintenance). The Total Annualised Cost over a 10-year plant life is:

$$TAC(N) = 15,000 \times N + 20,000 \quad (\text{Rs. /year})$$

(5.1)

Volume-dependent capital (Rs. 2,00,000) stays constant since total volume is fixed. Only the per-reactor fixed cost and operating cost scale with N.

5.2 Results: Conversion vs. Cost

Table 5.1: Performance for different N (green row = recommended optimum, N = 3)

N	X (%)	C _{A,out} (mg/L)	TAC (Rs./yr)	Extra X from prev.
1	83.33	16.67	35,000	--
2	91.84	8.16	50,000	8.51%
3*	94.73	5.27	65,000	2.89%
4	96.10	3.90	80,000	1.37%
5	96.88	3.13	95,000	0.78%
6	97.36	2.64	1,10,000	0.48%
10	98.27	1.73	1,70,000	--
PFR	99.33	0.67	--	--

5.3 Marginal Cost Analysis -- Why N = 3

Each extra reactor costs Rs. 15,000/year. But the conversion gain keeps shrinking.

Table 5.2: Cost of each additional reactor

Transition	Cost Added (Rs./yr)	Cost per 1% Gain (Rs.)	Worth It?
1st to 2nd	15,000	1,763	Absolutely
2nd to 3rd	15,000	5,190	Yes
3rd to 4th	15,000	10,949	Debatable
4th to 5th	15,000	19,231	No
5th to 6th	15,000	31,250	No

Cost per 1% gain doubles after $N = 3$. That is the knee point -- beyond this, you pay a lot more for very little improvement.

5.4 Optimisation Framework and Damkohler Number

The total cost is:

$$C_{total} = C_{capital}(N) + C_{operational}(N)$$

(5.2)

Capital cost increases with N , conversion benefit is asymptotic. The optimal N^* exists where $d(\text{Profit})/dN = 0$. The Damkohler number:

$$Da = \frac{kV}{F}$$

(5.3)

When $Da \ll 1$, the system is reaction-limited with little benefit from multiple reactors. When $Da \gg 1$, staging becomes beneficial. Wastewater systems operate at high Da .

Chapter 6

Environmental and Regulatory Aspects

6.1 Effluent Standards

EPA (Environmental Protection Agency) secondary treatment limits require: BOD₅ (Biochemical Oxygen Demand, 5-day) less than or equal to 30 mg/L, and TSS (Total Suspended Solids) less than or equal to 30 mg/L on a 30-day average. Some regions also require limits on TN (Total Nitrogen) and TP (Total Phosphorus). Disinfection (UV or chlorine) is often done after CSTR treatment.

6.2 Sludge Management

Biomass (sludge) produced must be treated and disposed. Methods include anaerobic digestion, drying, and disposal. Must follow biosolid regulations (Pathogen Class A/B standards).

6.3 Emissions

Aerobic CSTRs release CO₂ (from BOD oxidation). Anaerobic CSTRs produce CH₄ (methane, can be used as energy) and CO₂. Odour control required using covers and scrubbers.

6.4 Reuse of Treated Water

Can be reused for irrigation or industrial purposes. Must meet WHO / EPA guidelines, requiring low pathogens and controlled nutrient levels. Example: Fecal coliform less than 1000 per 100 mL for irrigation. CSTR + post-treatment must ensure these standards are met.

Chapter 7

Discussion

7.1 Why Does Performance Improve?

A single CSTR has complete back-mixing, which dilutes the incoming feed immediately and reduces the driving force for reaction. Multiple CSTRs in series reduce concentration gradients and create pseudo plug-flow behaviour. Each reactor operates at progressively lower reactant concentration, improving driving force utilisation.

7.2 Process Design Implications

Multiple CSTRs in series are used in activated sludge systems (wastewater) and multi-stage reactors in chemical plants. Benefits: better conversion control, flexibility in operation. Considerations: space constraints, maintenance complexity, energy consumption.

7.3 Limitations

This report relies on ideal assumptions: perfect mixing, steady-state, first-order kinetics. Ignored: heat effects leading to non-isothermal behaviour (except adiabatic section), RTD (Residence Time Distribution) effects, axial dispersion. Real reactors deviate from ideal CSTR/PFR models.

7.4 Future Scope

Future work: non-ideal flow modelling using RTD curves (E-curve, F-curve), CFD-based reactor simulation (e.g. ANSYS Fluent), AI/ML optimisation to predict optimal reactor networks, multi-objective optimisation (conversion + cost + energy).

Chapter 8

Conclusion

This project studied wastewater treatment using multiple CSTRs in series. The working equations for both isothermal and adiabatic configurations were presented, along with Monod kinetics and the ASM framework for biological modelling. The key findings are:

1. Conversion increases with the number of CSTRs. As N approaches infinity, the system approaches the PFR limit. The performance improvement is non-linear -- marginal gain dX/dN decreases as N increases.
2. The adiabatic reactor is 2-5 times more efficient in volume than isothermal for exothermic reactions. However, isothermal operation is preferred for biological wastewater treatment because microbes require controlled temperature.
3. The cost model shows $N = 3$ is optimal. At $N = 3$, we achieve 94.73% conversion at Rs. 65,000/year. Cost per 1% gain doubles after this point.
4. Aerobic CSTRs suit BOD/TSS removal (faster, higher energy). Anaerobic CSTRs handle high-COD waste and produce biogas but cannot remove nutrients.
5. At high Da (typical for wastewater), staging is beneficial, but diminishing returns set in quickly beyond 3 reactors.

Reactor design is an optimisation problem, not a maximisation problem. The engineer's job is to find the design point that best balances cost, efficiency, and safety. For the wastewater treatment system studied here, that point is $N = 3$.